Introduction The integration of robotic manipulators into electronics manufacturing has traditionally relied on high-cost, rigid industrial systems. However, the rise of agile micro-factories requires adaptable, low-cost manipulators for secondary quality control tasks, such as Printed Circuit Board (PCB) handling. This project proposes the simulation and kinematic analysis of a 6-Degree-of-Freedom (DOF) low-cost robotic arm to perform automated PCB defect sorting at a discrete inspection station on an indexing conveyor belt.

Literature Review Electronics manufacturing demands high precision, particularly during the Surface-Mount Technology (SMT) assembly process. After reflow soldering, boards undergo Automated Optical Inspection (AOI) to detect defects such as missing components, solder bridges, or tombstoning. Traditionally, sorting defective PCBs from an assembly line has relied on manual labor or expensive, heavy-duty arms (e.g., UR5e). However, lightweight, open-source manipulators driven by smart servos offer sufficient repeatability for light payloads like PCBs, significantly lowering the barrier to entry for automated sorting cells.

To ensure stability during fragile board handling, modern AOI and rework stations often utilize an indexing (step-and-dwell) conveyor system. Rather than tracking a continuously moving target, the conveyor discretely advances the PCB into an inspection nest and pauses. During this dwell period, a vision system classifies the board's status, and the robotic manipulator computes the Inverse Kinematics (IK) to grasp the PCB. To simplify the computer vision pipeline and isolate the kinematic analysis, the optical properties of the simulation are deliberately constrained; the colors of the PCBs and their corresponding sorting bins are hardcoded. This approach eliminates the need for complex feature extraction, ensuring reliable state detection based strictly on color thresholding.

The 6-DOF configuration of the proposed low-cost arm provides a distinct kinematic advantage over 4-DOF and 5-DOF underactuated models. It allows for arbitrary end-effector orientation, ensuring a perfectly perpendicular approach vector to the flat PCB surface. By simulating this indexed workflow in MuJoCo integrated with ROS2, this project will validate complex task-space trajectories, velocity/acceleration kinematics, and motor-level control algorithms required to sort boards into "Pass," "Rework," or "Scrap" bins before physical deployment.